Benchmark: **MA.7.NSO.1.2** · Rewrite rational numbers in equivalent forms Student code: _____ Calculator: no

Learning goal: I can rewrite rational numbers in equivalent forms. (circle your level after scoring)

4	I can choose the best form of a number to solve a problem and explain why it is equivalent.
3	I can convert between fractions, decimals, and percents, including repeating decimals.
2	I can convert between a fraction and a terminating decimal.
1	I need help to start this benchmark.

1. Write the repeating decimal as a fraction in simplest form.

$0.\overline{46}$

2. Write the fraction as a repeating decimal.

$\frac{13}{33}$

Turn over →

3. Two identical bottles of sports drink are partly full. The first bottle is $\frac{10}{11}$ full and the second bottle is 95% full.

The first bottle holds ____ the second bottle.

Choose the phrase that correctly completes the sentence.

A. exactly twice as much as

B. the same amount as

C. more than

D. less than

4. Write the fraction as an exact percent.

$\frac{7}{11}$

5. Maya uses a calculator to find $\frac{190}{33}$. The calculator shows 5.757575758. Maya says that $\frac{190}{33} = 5.757575758$.

Is Maya correct? Explain, and write the exact value of $\frac{190}{33}$ as a repeating decimal.

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-NSO.1.2 · MA.7.NSO.1.2 · Rewrite rational numbers in equivalent forms

Unit 2: Probability. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular $\frac{46}{99}$

Two repeating digits go over 99: $0.\overline{46} = \frac{46}{99}$.

2. Regular $0.\overline{39}$

$13 \div 33 = 0.393939\ldots$, so $\frac{13}{33} = 0.\overline{39}$: the digits 39 repeat.

3. Regular **D. less than**

Rewrite both amounts in the same form. $\frac{10}{11} = 10 \div 11$, which is about 90.91%.

90.91% is less than 95%, so the first bottle holds less than the second bottle.

4. Regular $63\frac{7}{11} = \frac{700}{11}\%(\text{or } 63.\overline{63}\%)$

A percent is the value out of 100: $7 \div 11 \times 100 = 63\frac{7}{11}\%$, which is $63.\overline{63}\%$ as a repeating decimal.

5. Rigor **No. The calculator rounded a repeating decimal, so 5.757575758 is not exact; $\frac{190}{33} = 5\frac{25}{33} = 5.\overline{75}$. Score: 1 point for "No" with the rounding reason and the exact value $5.\overline{75}$ (or an equivalent repeating form); "No" alone earns 0.**

Divide: $190 \div 33 = 5.757575\ldots$, and the digits 75 repeat forever.

A calculator shows only so many digits and rounds the last one, so 5.757575758 is close to $\frac{190}{33}$ but not equal to it.

The exact value is $5.\overline{75}$.

Re-check strip (3 items, C-NSO.1.2-R)

1. Regular **B. more than**

Rewrite both amounts in the same form. $\frac{4}{11} = 4 \div 11$, which is about 36.36%.

36.36% is more than 30%, so the first bottle holds more than the second bottle.

2. Regular **A.** $0.\overline{08}$**; C.** $\frac{16}{198}$

$8 \div 99 = \frac{8}{99} = 0.\overline{08}$ (the block 08 repeats forever).

$\frac{8}{99} = \frac{16}{198}$, so $\frac{16}{198}$ also names the quotient.

0.8 repeats only the digit 8, not 08.

$\frac{99}{8}$ is $99 \div 8$, not $8 \div 99$.

0.0808 stops after 4 decimal places; it is not the repeating decimal.

Scoring: 1 point only if every correct box is checked and no other box is; anything else earns 0.

3. Regular $77\frac{7}{9} = \frac{700}{9}\%(\text{or } 77.\overline{7}\%)$

A percent is the value out of 100: $7 \div 9 \times 100 = 77\frac{7}{9}\%$, which is $77.\overline{7}\%$ as a repeating decimal.

Comparison strip (2 items, C-NSO.1.2-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **A. less than**

Rewrite both amounts in the same form. $\frac{2}{15} = 2 \div 15$, which is about 13.33%.

13.33% is less than 15%, so the first bottle holds less than the second bottle.

2. Regular **B.** $\frac{3}{99}$**; C.** $0.\overline{03}$

$1 \div 33 = \frac{1}{33} = 0.\overline{03}$ (the block 03 repeats forever).

$\frac{1}{33} = \frac{3}{99}$, so $\frac{3}{99}$ also names the quotient.

0.0303 stops after 4 decimal places; it is not the repeating decimal.

$\frac{33}{1}$ is $33 \div 1$, not $1 \div 33$.

$0.\overline{3}$ repeats only the digit 3, not 03.

Scoring: 1 point only if every correct box is checked and no other box is; anything else earns 0.

Reteach plan — C-NSO.1.2

MA.7.NSO.1.2 · Rewrite rational numbers in equivalent forms

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-NSO.1.2-R

Benchmark: **MA.7.NSO.1.2** · Rewrite rational numbers in equivalent forms Student code: _____ Calculator: no

1. Two identical bottles of sports drink are partly full. The first bottle is $\frac{4}{11}$ full and the second bottle is 30% full.

The first bottle holds ____ the second bottle.

Choose the phrase that correctly completes the sentence.

A. the same amount as

B. more than

C. exactly twice as much as

D. less than

2. Select **all** the values that are equal to the quotient $8 \div 99$.

Select **all** the correct answers.

A. ☐ $0.\overline{08}$

B. ☐ $0.\overline{8}$

C. ☐ $\frac{16}{198}$

D. ☐ 0.0808

E. ☐ $\frac{99}{8}$

3. Write the fraction as an exact percent.

$\frac{7}{9}$

Score: ____ / 3 → copy to your progress strip

7th Grade Math · Comparison items (2) C-NSO.1.2-X

Benchmark: **MA.7.NSO.1.2** · Rewrite rational numbers in equivalent forms Student code: _____ Calculator: no

1. Two identical bottles of sports drink are partly full. The first bottle is $\frac{2}{15}$ full and the second bottle is 15% full.

The first bottle holds ____ the second bottle.

Choose the phrase that correctly completes the sentence.

A. less than

B. more than

C. the same amount as

D. exactly twice as much as

2. Select **all** the values that are equivalent to $1 \div 33$.

Select **all** the correct answers.

A. ☐ $\frac{33}{1}$

B. ☐ $\frac{3}{99}$

C. ☐ $0.\overline{03}$

D. ☐ 0.0303

E. ☐ $0.\overline{3}$

Score: ____ **/ 2** → **copy to your progress strip**

7th Grade Math · Comparison items (2) C-NSO.1.2-X

Benchmark: **MA.7.NSO.1.2** · Rewrite rational numbers in equivalent forms Student code: _____ Calculator: no

1. Two identical bottles of sports drink are partly full. The first bottle is $\frac{2}{15}$ full and the second bottle is 15% full.

The first bottle holds ____ the second bottle.

Choose the phrase that correctly completes the sentence.

A. less than

B. more than

C. the same amount as

D. exactly twice as much as

2. Select **all** the values that are equivalent to $1 \div 33$.

Select **all** the correct answers.

A. ☐ $\frac{33}{1}$

B. ☐ $\frac{3}{99}$

C. ☐ $0.\overline{03}$

D. ☐ 0.0303

E. ☐ $0.\overline{3}$

Score: ____ **/ 2** → **copy to your progress strip**